Qualification Pack

Industry 4.0

QP Code: ASC/Q8317

Version: 1.0

NSQF Level: 4.5

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Contents

ASC/Q8317: Industry 4.0

Brief Job Description

The role of the Digital Production Systems Technician is to understand the business case for enhancement, and to design and implement technical responses accordingly. Assembled and commissioned hardware in virtual and real context using various digital tools and technology provide the basis for programming, and the design and implementation of cyber security measures on real and virtual production processes.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ASC/N9831: Work organization and management (Industry 4.0)
2. ASC/N9832: Communication and interpersonal skills (Industry 4.0)
3. ASC/N8373: Design, assembly, and commissioning
4. ASC/N8374: Software design and implementation
5. ASC/N8375: Networking and cyber security
6. ASC/N8376: Testing, maintenance, and fault-finding
7. ASC/N8377: Enhancement and optimization
8. ASC/N8378: Analysis, evaluation, and reporting

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
Country	India

NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	25 Years
Last Reviewed On	NA
Next Review Date	08/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-AU-01820-2024-V1-ASDC
NQR Version	1

ASC/N9831: Work organization and management (Industry 4.0)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Manage work and resources

Elements and Performance Criteria

Manage work and resources

To be competent, the user/individual on the job must be able to:

PC1. Set up and maintain a safe, clean and efficient work area

PC2. Maintain an appropriate state of preparation and readiness to receive, schedule and act on requests and assignments efficiently, effectively and safely

PC3. Order, select, use and care for all equipment, facilities and materials in accordance with manufacturers' instructions and accepted good practice

PC4. Conduct self and all operations with care and consideration for other personnel, cost efficiency and the environment

PC5. Monitor progress, modifying or changing plans or approaches through a rational process, within their personal authority

PC6. Complete assignments or tasks, and restore the work area to its state of readiness for future use

PC7. Reflect on and review their personal performance, as part of continuing professional development

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The principles and parameters of integrated automated production

KU2. Their specific roles within integrated automated production

KU3. Principles, applications, accountabilities and techniques for project management

KU4. Principles and applications of safe working practice broadly and specifically

KU5. The purpose, use, care and maintenance of equipment, facilities and materials

KU6. Principles and methods for organizing, controlling and managing work and its outcomes

KU7. Their personal strengths and limitations relative to the roles, projects and tasks assigned

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Manage work and resources	**15**	**25**	**-**	**10**
PC1. Set up and maintain a safe, clean and efficient work area	2	3	-	1
PC2. Maintain an appropriate state of preparation and readiness to receive, schedule and act on requests and assignments efficiently, effectively and safely	2	3	-	1
PC3. Order, select, use and care for all equipment, facilities and materials in accordance with manufacturers' instructions and accepted good practice	2	4	-	2
PC4. Conduct self and all operations with care and consideration for other personnel, cost efficiency and the environment	2	4	-	2
PC5. Monitor progress, modifying or changing plans or approaches through a rational process, within their personal authority	2	3	-	1
PC6. Complete assignments or tasks, and restore the work area to its state of readiness for future use	3	5	-	2
PC7. Reflect on and review their personal performance, as part of continuing professional development	2	3	-	1
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9831
NOS Name	Work organization and management (Industry 4.0)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N9832: Communication and interpersonal skills (Industry 4.0)

Description

This unit is about communicating with team members, superior and others.

Scope

The scope covers the following :

- Communicate effectively with others

Elements and Performance Criteria

Communicate effectively with others

To be competent, the user/individual on the job must be able to:

PC1. Receive assignments, identify their salient points, and ask questions for clarification and confirmation

PC2. Read, interpret and extract technical data and instructions from given documentation in all available formats

PC3. Discuss and plan with relevant others the complex, joint and overlapping elements of assignments

PC4. Communicate verbally, in writing, and electronically, using methods that ensure clarity, efficiency and effectiveness

PC5. Make and retain reports on progress, issues and actions, in the required formats

PC6. Give and take feedback and support to and from others

PC7. Review the team’s performance, one’s own con4tribution, and individual and collective learning points

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Their personal strengths and limitations: In perception and awareness, In communication with known and unknown others, In working as a colleague, leader, learner or assistant

KU2. Principles of communication and purposeful social learning

KU3. Standards and protocols for formal and informal, direct and indirect communication with team members, managers and clients

KU4. The technical language required for the role, including the content and structures of the English language

KU5. Standards and protocols for communicating electronically and in the cyber space

KU6. The scope and purposes of documentation in hard copy and electronic format

KU7. The requirements for routine and exception reports, in all formats

KU8. Principles and methods for analysing, synthesizing, using and communicating data

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Communicate effectively with others	**15**	**25**	**-**	**10**
PC1. Receive assignments, identify their salient points, and ask questions for clarification and confirmation	2	3	-	1
PC2. Read, interpret and extract technical data and instructions from given documentation in all available formats	2	3	-	1
PC3. Discuss and plan with relevant others the complex, joint and overlapping elements of assignments	2	4	-	2
PC4. Communicate verbally, in writing, and electronically, using methods that ensure clarity, efficiency and effectiveness	2	4	-	2
PC5. Make and retain reports on progress, issues and actions, in the required formats	2	3	-	1
PC6. Give and take feedback and support to and from others	3	5	-	2
PC7. Review the team's performance, one's own con4tribution, and individual and collective learning points	2	3	-	1
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9832
NOS Name	Communication and interpersonal skills (Industry 4.0)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8373: Design, assembly, and commissioning

Description

This NOS is about designing, assembling, and commissioning of the components of system.

Scope

The scope covers the following :

- Design, assembly, and commissioning of system

Elements and Performance Criteria

Design, assembly, and commissioning of system

To be competent, the user/individual on the job must be able to:

PC1. Read and interpret instructions, using questioning techniques and research to check, verify and prepare

PC2. Design systems for the automation and communication of production modules, with the given parameters for cyber-physical systems

PC3. Test and implement design solutions

PC4. Assemble machines and equipment

PC5. Select and apply sensors, communication technologies, and devices for motion control, position sensing, pressure testing and electronic communication

PC6. Test the performance of electrical, electronic, mechanical and integrated systems and equipment, relative to their intended purpose

PC7. Apply mechatronic or automated solutions to the transfer of materials, components or finished goods

PC8. Integrate the equipment and sub-systems to ensure readiness for data capture, networking, exchange and use

PC9. Commission the system

PC10. Create and maintain project files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Practical applications of engineering science and technology to the design and production of goods and services in virtual and real context

KU2. Principles and directions for integrating local/artificial intelligence with wider communication capacities

KU3. Principles and applications for the Design, Assembly, Connectivity and Commissioning of hardware and peripherals to meet cyber-physical requirements

KU4. Principles and methods for integrating autonomous subsystems and components

KU5. Principles and applications for data collection, storage, networking and use

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Design, assembly, and commissioning of system	**30**	**50**	**-**	**20**
PC1. Read and interpret instructions, using questioning techniques and research to check, verify and prepare	3	5	-	2
PC2. Design systems for the automation and communication of production modules, with the given parameters for cyber-physical systems	3	5	-	2
PC3. Test and implement design solutions	3	5	-	2
PC4. Assemble machines and equipment	3	5	-	2
PC5. Select and apply sensors, communication technologies, and devices for motion control, position sensing, pressure testing and electronic communication	3	5	-	2
PC6. Test the performance of electrical, electronic, mechanical and integrated systems and equipment, relative to their intended purpose	3	5	-	2
PC7. Apply mechatronic or automated solutions to the transfer of materials, components or finished goods	3	5	-	2
PC8. Integrate the equipment and sub-systems to ensure readiness for data capture, networking, exchange and use	3	5	-	2
PC9. Commission the system	3	5	-	2
PC10. Create and maintain project files	3	5	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8373
NOS Name	Design, assembly, and commissioning
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8374: Software design and implementation

Description

The NOS is about designing and implementation of the software for the system.

Scope

The scope covers the following :

- Designing and implementation of software

Elements and Performance Criteria

Designing and implementation of software

To be competent, the user/individual on the job must be able to:

PC1. Write, analyse, review, and rewrite programs

PC2. Correct errors by making appropriate changes and rechecking that the desired results are produced

PC3. Perform or direct revision, repair, or expansion of existing programs to increase operating efficiency or adapt to new requirements

PC4. Write, update, and maintain computer programs or software packages to handle specific jobs such as tracking inventory, storing or retrieving data, or controlling other equipment

PC5. Conduct trial runs of programs and software applications to ensure they produce the desired information and the instructions are correct

PC6. Prepare detailed workflow charts and diagrams that describe input, output, and logical operation, and convert them into a series of instructions coded in a computer language

PC7. Compile and write documentation of program development and subsequent revisions, using protocols to ensure that others can understand the programs

PC8. Consult with others to define and resolve problems in running programs

PC9. Perform systems analysis and programming modules to maintain and control the use of computer systems software

PC10. Write or contribute to instructions or manuals to guide end users

PC11. Investigate whether networks, workstations, the central processing unit of the system, or peripheral equipment are responding to a program's instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Mathematics and their applications

KU2. Principles and applications of electronics

KU3. Computer capabilities, subject matter, and symbolic logic

KU4. Computer hardware and software, and their applications

KU5. The required standards for code conventions, style guides, user interface designs, managing directories, and files

KU6. Principles and applications of human-machine communication

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Designing and implementation of software	**30**	**50**	**-**	**20**
PC1. Write, analyse, review, and rewrite programs	3	5	-	2
PC2. Correct errors by making appropriate changes and rechecking that the desired results are produced	3	5	-	2
PC3. Perform or direct revision, repair, or expansion of existing programs to increase operating efficiency or adapt to new requirements	3	5	-	2
PC4. Write, update, and maintain computer programs or software packages to handle specific jobs such as tracking inventory, storing or retrieving data, or controlling other equipment	3	5	-	2
PC5. Conduct trial runs of programs and software applications to ensure they produce the desired information and the instructions are correct	3	5	-	2
PC6. Prepare detailed workflow charts and diagrams that describe input, output, and logical operation, and convert them into a series of instructions coded in a computer language	3	5	-	2
PC7. Compile and write documentation of program development and subsequent revisions, using protocols to ensure that others can understand the programs	3	5	-	2
PC8. Consult with others to define and resolve problems in running programs	2	3	-	2
PC9. Perform systems analysis and programming modules to maintain and control the use of computer systems software	3	5	-	2
PC10. Write or contribute to instructions or manuals to guide end users	2	3	-	1
PC11. Investigate whether networks, workstations, the central processing unit of the system, or peripheral equipment are responding to a program's instructions	2	4	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8374
NOS Name	Software design and implementation
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8375: Networking and cyber security

Description

This NOS is about carrying out activities related to networking and cyber security of the system.

Scope

The scope covers the following :

- Networking and cyber security activities

Elements and Performance Criteria

Networking and cyber security activities

To be competent, the user/individual on the job must be able to:

PC1. Design and implement network protocols and topologies

PC2. Develop plans to safeguard computer files against accidental or unauthorized modification, destruction, or disclosure, and meet emergency data processing needs

PC3. Maintain levels of preparedness and the availability of preventative and defensive tools commensurate with risks and trends in malicious attacks

PC4. Monitor reports of computer viruses to determine when to update virus protection systems

PC5. Encrypt data transmissions and erect firewalls to conceal confidential information during transmitted, and to keep out tainted digital transfers

PC6. Perform risk assessments and conduct tests of data processing systems to ensure safe functioning of data processing and security measures

PC7. Modify computer security files to incorporate new software, correct errors, or change individual access status

PC8. Monitor the use of data files and regulate access to safeguard information

PC9. Review violations of procedures and take steps to prevent their repeating

PC10. Document computer security and emergency measures, policies, procedures and tests

PC11. Test and simulate disaster recovery plans

PC12. Train users and promote security awareness to ensure system security and improve server and network efficiency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The scale and nature of the organization vulnerability to breaches in information security

KU2. The trends, nature and intent of malicious breaches

KU3. The nature and causes of incidental and accidental data breaches, both human and systemic

KU4. Principles and methodologies for establishing and maintaining maximum information security and data integrity

KU5. Principles and methodologies for addressing minor breaches
KU6. Principles for the design and execution of disaster recovery plans
KU7. Development environment software
KU8. Network protocols and topology
KU9. Network monitoring software
KU10. Transaction security and virus protection software
KU11. Web platform development software

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Networking and cyber security activities	**30**	**50**	**-**	**20**
PC1. Design and implement network protocols and topologies	3	5	-	2
PC2. Develop plans to safeguard computer files against accidental or unauthorized modification, destruction, or disclosure, and meet emergency data processing needs	3	5	-	2
PC3. Maintain levels of preparedness and the availability of preventative and defensive tools commensurate with risks and trends in malicious attacks	3	5	-	2
PC4. Monitor reports of computer viruses to determine when to update virus protection systems	3	5	-	2
PC5. Encrypt data transmissions and erect firewalls to conceal confidential information during transmitted, and to keep out tainted digital transfers	3	5	-	2
PC6. Perform risk assessments and conduct tests of data processing systems to ensure safe functioning of data processing and security measures	3	5	-	2
PC7. Modify computer security files to incorporate new software, correct errors, or change individual access status	3	5	-	1
PC8. Monitor the use of data files and regulate access to safeguard information	2	3	-	2
PC9. Review violations of procedures and take steps to prevent their repeating	2	3	-	1
PC10. Document computer security and emergency measures, policies, procedures and tests	1	2	-	1
PC11. Test and simulate disaster recovery plans	3	5	-	2

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Train users and promote security awareness to ensure system security and improve server and network efficiency	1	2	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8375
NOS Name	Networking and cyber security
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8376: Testing, maintenance, and fault-finding

Description

This NOS is about testing, maintenance, and fault-finding of the system.

Scope

The scope covers the following :

- Testing, maintenance, and fault-finding activities

Elements and Performance Criteria

Testing, maintenance, and fault-finding activities

To be competent, the user/individual on the job must be able to:

PC1. Identify the parts of the production system to which to apply smart maintenance

PC2. Establish the parameters for the parts' operation

PC3. Use the access tools at the appropriate data points, or on a mobile basis

PC4. Monitor the condition of each part, using augmented reality or other tools as helpful

PC5. Discuss and check findings with relevant personnel

PC6. Undertake preventive or predictive maintenance by reviewing alternative courses of action and scheduling or recommending the optimal measure(s)

PC7. Use the available technology and measures to effect maintenance with least disruption to production

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Principles and applications of smart maintenance, based on data, to enable :Condition monitoring, Data analysis and correlation, Predictive maintenance, Mobile maintenance

KU2. The use of augmented reality and other emerging technologies and tools

KU3. The use of simulation models, reconfiguration and virtualization

KU4. Operational parameters/process data

KU5. The use of constraints and variables, restrictions, alternatives,conflicting objectives, and numerical parameters for conceptualizing and defining problems

KU6. Principles and methodologies for designing alternatives and making decisions and recommendations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Testing, maintenance, and fault-finding activities	**15**	**25**	**-**	**10**
PC1. Identify the parts of the production system to which to apply smart maintenance	2	3	-	1
PC2. Establish the parameters for the parts' operation	2	3	-	1
PC3. Use the access tools at the appropriate data points, or on a mobile basis	2	4	-	2
PC4. Monitor the condition of each part, using augmented reality or other tools as helpful	2	4	-	2
PC5. Discuss and check findings with relevant personnel	2	3	-	1
PC6. Undertake preventive or predictive maintenance by reviewing alternative courses of action and scheduling or recommending the optimal measure(s)	3	5	-	2
PC7. Use the available technology and measures to effect maintenance with least disruption to production	2	3	-	1
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8376
NOS Name	Testing, maintenance, and fault-finding
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8377: Enhancement and optimization

Description

This NOS is about enhancement and optimization of the system.

Scope

The scope covers the following :

- Enhancement and optimization of system

Elements and Performance Criteria

Enhancement and optimization of system

To be competent, the user/individual on the job must be able to:

PC1. Reduce costs by removing waste ad consumption caused by: Over-production, Stock and storage, Over- and unnecessary processing, Poor quality, Transport and movement, Waiting time

PC2. Analyse and recommend opportunities for optimization using: Simulations, Prototyping, Digital shadows/twins

PC3. Identify opportunities for: Greater lateral and vertical integration, The use of the Cloud

PC4. Identify the cost-benefit implications, financial and human, of optimization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The potential for smart production systems to be enhanced to: Enable greater flexibility and individualization in production, Shorten reaction and response time in production, Reduce time and cost in production, Collect, share and use information for continuous enhancement

KU2. Principles and methods for identifying, analysing and pursuing opportunities for enhancement

KU3. The implications of increased data storage and exchange

KU4. Principles and methods for cost benefit analysis

KU5. Principles and methods for work organization and workforce planning and development

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Enhancement and optimization of system	**15**	**25**	**-**	**10**
PC1. Reduce costs by removing waste ad consumption caused by: Over-production, Stock and storage, Over- and unnecessary processing, Poor quality, Transport and movement, Waiting time	3	6	-	2
PC2. Analyse and recommend opportunities for optimization using: Simulations, Prototyping, Digital shadows/twins	4	7	-	3
PC3. Identify opportunities for: Greater lateral and vertical integration, The use of the Cloud	4	6	-	2
PC4. Identify the cost-benefit implications, financial and human, of optimization	4	6	-	3
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8377
NOS Name	Enhancement and optimization
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N8378: Analysis, evaluation, and reporting

Description

This NOS is about performing analysis and evaluation of the system performance and do the reporting to the concerned person

Scope

The scope covers the following :

- Analysis, evaluation and reporting activities

Elements and Performance Criteria

Analysis, evaluation and reporting activities

To be competent, the user/individual on the job must be able to:

PC1. Take account of requirements for monitoring, review and evaluation in the design of the system and sub-systems

PC2. Optimize the use of self-monitoring equipment and tools to the extent feasible

PC3. Design and apply an appropriate model for monitoring and evaluating performance relative to specification

PC4. Anticipate requests for feedback and reports, and prepare accordingly on a data rational basis

PC5. Prepare reports in appropriate formats for routine and exception reporting

PC6. Make presentations customized to particular groups and individuals

PC7. Maintain awareness of new possibilities and options for improvement, making recommendations on the basis of return on investment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Principles and applications of critical thinking and complex problem-solving

KU2. The uses and availability of self-monitoring equipment and tools

KU3. The bases, techniques and tools for creating and using analytical

KU4. Models of performance, including: Performance targets or specifications, Numerical and quantifiable parameters, Data requirements, Constraints and variables, Alternatives

KU5. How to conceptualize, define and evaluate problems referred to them, and to derive recommendations for solutions

KU6. The content, structure and presentation for reports serving different purposes

KU7. Principles and applications to presentations for management, peers and clients

KU8. Cost benefit analysis, and its uses for recommending alternative courses of action

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Analysis, evaluation and reporting activities	**15**	**25**	**-**	**10**
PC1. Take account of requirements for monitoring, review and evaluation in the design of the system and sub-systems	2	3	-	1
PC2. Optimize the use of self-monitoring equipment and tools to the extent feasible	2	3	-	1
PC3. Design and apply an appropriate model for monitoring and evaluating performance relative to specification	2	4	-	2
PC4. Anticipate requests for feedback and reports, and prepare accordingly on a data rational basis	2	4	-	2
PC5. Prepare reports in appropriate formats for routine and exception reporting	2	3	-	1
PC6. Make presentations customized to particular groups and individuals	3	5	-	2
PC7. Maintain awareness of new possibilities and options for improvement, making recommendations on the basis of return on investment	2	3	-	1
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8378
NOS Name	Analysis, evaluation, and reporting
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9831.Work organization and management (Industry 4.0)	15	25	0	10	50	5
ASC/N9832.Communication and interpersonal skills (Industry 4.0)	15	25	0	10	50	5
ASC/N8373.Design, assembly, and commissioning	30	50	0	20	100	20
ASC/N8374.Software design and implementation	30	50	0	20	100	20
ASC/N8375.Networking and cyber security	30	50	0	20	100	20
ASC/N8376.Testing, maintenance, and fault-finding	15	25	0	10	50	15
ASC/N8377.Enhancement and optimization	15	25	0	10	50	10
ASC/N8378.Analysis, evaluation, and reporting	15	25	0	10	50	5
Total	**165**	**275**	**0**	**110**	**550**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today’s world. These skills are typically needed in any work environment in today’s world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.